P3: PROJECT PLANNING • TEAM REPOSITORY: /PLAN

Lost And Found System (LAF System)

Project Planning Report: Tasks 1 to 5 (Process Justification, Estimation, Schedule, Risk Register, Plan Skeleton)

Task 1: Software Process Model Justification (Scrum)

Scrum is the most appropriate software process model for the LAF SYSTEM because requirements are likely to change as user feedback on lost item reporting and claim validation emerges, regular feedback is received from our supervisor/product owner, our team is small (4 members), and progress stays visible every sprint through working software increments. The trade-off we accept is having less upfront documentation, which we cover through structured milestone reports (M1 to M3) plus a fixed outer schedule for the Week 7 (Part A) and Week 13 (Part B) deadlines.

Task 2: Effort Estimation & Capacity Check

1. First Pass Estimates (Person-Days)

#	Backlog Work Item	Best	Expected	Worst
1	User authentication and National ID registration	3	5	8
2	Password reset and customer service	2	3	7
3	Lost and found item reporting	3	5	8
4	Search and filtering	2	4	6
5	Auto-generated item ID tags	1	2	5
6	Owner verification and ownership proof	2	4	7
7	Find/owner contact	2	3	6
8	Reviews	1	2	4
9	UI/UX design	3	5	9
First Pass Total		**19**	**33**	**60**

Group estimation note: Item 9 (UI/UX design) had the biggest gap (Zhima said 3, Nasik said 9). Zhima estimated 3 days because she only thought of the basic layout and standard CSS templates, however Nasik estimated 9 days because she included creating custom designs in Figma, making them work on mobile phones, and improving accessibility and ease of use. After discussing the work needed, the team agreed on 5 person-days.

Initial Contingency & Capacity Check:
• **Contingency:** 33 × 1.4 = 46.2 ≈ 46 person-days (with 40% contingency).
• **Capacity Check:** 4 people × 10 weeks (Week 4 to Week 13) × 6 hours = 240 hours = 40 person-days. 46 > 40, so it does not fit.
• **Scope Cuts (lowest-priority backlog items first):**

- Drop Reviews (saves 2 person-days).
- Simplify Search and Filtering to basic keyword search (4 down to 2, saves 2 person-days).
- Password reset and customer service reduced by 1 (saves 1 person-day).

• **Revised Total:** 15 / 28 / 53 person-days. With 40% contingency: 28 × 1.4 = 39.2 ≈ 39 person-days, which fits within the 40 person-day capacity with 1 spare day.

Task 3: Project Plan and Schedule

1. Task Table (Effort in person-days, duration in weeks)

Task	Description	Effort	Duration	Depends on	Owner	Weeks	Critical
T1	User authentication and National ID registration	5	2	None	Nasik	W4–5	Yes
T2	Password reset and customer service	2	1	T1	Ayoub	W6	No (2 wks float)
T3	Lost and found item reporting	5	2	T1	Zhima, Ayoub	W6–7	Yes
T4	Search and filtering	2	1	T3	Dareen	W7	No (1 wk float)
T5	Auto-generated item ID tags	2	1	T3	Ayoub, Zhima	W8	Yes
T6	Owner verification and ownership proof	4	2	T3, T5	Dareen, Ayoub	W8–9	Yes
T7	Find/owner contact	3	1	T6	Zhima	W10	Yes
T8	UI/UX design	5	3	None	Nasik, Ayoub	W4–6	No (parallel)
T9	Part A	2	1	T1, T3	Nasik	W7	Fixed Deadline
T10	Testing and Bug Fixing	2	1	T2, T4, T7	All	W11–12	Yes
T11	Final Documentation and Presentation	2	1	T10	All	W13	Yes
Total Effort		**34 person-days**					

2. Milestones and Deliverables

• **Milestones (short reviews):** M1 (end Week 5): User authentication and National ID registration; M2 (end Week 10): Find/owner contact; M3 (end Week 12): Testing and Bug Fixing.
• **Deliverables:** Part A report (end Week 7); Final Documentation and Presentation (Week 13).

3. Critical Path & Slack Analysis

• **Critical path:** T1 → T3 → T5 → T6 → T7 → T10 → T11 = 2 + 2 + 1 + 2 + 1 + 1 + 1 = 10 weeks (exactly Weeks 4 to 13). Strictly critical with zero slack.
• **Float/Slack:** T2 has 2 weeks float (can slip to Week 8). T4 has 1 week float (can slip to Week 8). T8 runs in parallel across Weeks 4–6. T9 has a hard deadline at Week 7.

4. Rough Bar Chart (Schedule Matrix)

Task / Week	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13
T1 (Nasik)										
T2 (Ayoub)										
T3 (Zhima, Ayoub)										
T4 (Dareen)										
T5 (Ayoub, Zhima)										
T6 (Dareen, Ayoub)										
T7 (Zhima)										
T8 (Nasik, Ayoub)										
T9 (Nasik)										
T10 (All)										
T11 (All)										

Legend: Dark blue = Critical path task; Light blue = Non-critical / parallel task with float; Dark gray = Fixed milestone deliverable.

5. Sprint 1 Plan (2 weeks, Weeks 4–5)

• **Sprint Goal:** Build a working login and registration system and prepare the basic UI designs.
• **Scope & Capacity:** The team will work on T1 (5 person-days), T8 (5 person-days), setting up the framework, matching the design system, and configuring team tools. The team plans to use about half of its total working capacity because initial velocity is unknown. Progress will be reviewed at the end of Week 5 to recalibrate actual velocity.

Task 4: Risk Register

1. Brainstormed and Rated Risks (Sorted by Seriousness)

ID	Risk	Type	Class	Probability	Effect
R1	Development time underestimated (worst case 42 person-days vs capacity 40, no slack)	Estimation	Project	High	Serious
R2	Exams and other deadlines reduce availability in Weeks 8 to 11	Organisational	Project	High	Serious
R3	Unfamiliar framework and secure ID hashing slow T1, on critical path	Technology	Project	High	Serious
R4	Supervisor/stakeholder changes proof-of-ownership rules after Part A feedback	Requirements	Product	Moderate	Serious
R5	Free hosting or tool limits block deployment or image uploads	Tools	Project	Moderate	Tolerable
R6	A team member leaves or the team has a serious conflict / illness	People	Project	Low	Serious

2. Key Risks Selected

The team kept R1, R2 and R3 because each has high probability and serious effect. R4 and R6 stay on the list but are not planned in detail yet. R5 is only watched.

3. Plans for the Key Risks

ID	Strategy Type	Plan	Early-Warning Indicator	Owner
R1	Minimisation, plus a contingency plan	40% contingency is already in the estimate. Track actual vs planned effort weekly. If behind, cut in this order: user reviews, contact system extras, list filtering.	Less than 70% of the sprint's planned work done, or velocity below 3 person-days a week	Ayoub
R2	Avoidance	Collect all exam dates in Week 4 and put them in the schedule. Finish T5 by Week 10 as planned. Use the spare capacity in Weeks 9 and 10.	Members report fewer hours than planned at the weekly meeting	Dareen
R3	Avoidance, plus a contingency plan	Spend the first days of Sprint 1 on a small working endpoint (a spike) with ID hashing. Pair on T1. If the spike fails by end of Week 4, switch to the simpler stack members already know.	No working endpoint by the end of Week 4	Zhima

4. Monitoring

- Review R1 to R3 at every weekly meeting.
- Review the whole register in each sprint retrospective and at every plan review (every 2 to 3 weeks).
- Add or retire risks when something changes, for example R2 drops once exams are over.

Task 5: Project Plan Skeleton

One-Page Plan Skeleton: Lost And Found System (LAF System)

Team	Ayoub Ahmed, Dareen Ahmed, Nasik Fouad, Zhima Karkhi
Version and date	v0.1, Week 4 (October 2026)
Plan location	Team shared repository, folder /plan
First plan review	End of Week 6 (then every 2 to 3 weeks)

The Seven Headings (Section 23.2.1)

#	Heading	One-Line Entry	Status
1	**Introduction**	Build a web app where students report and find lost items; fixed deadlines are Part A (end of Week 7) and Part B (Week 13); scope is 28 person-days within a capacity of 40.	**Complete**
2	**Project organisation**	Scrum with 2-week sprints. Ayoub is Scrum Master, Zhima is lead developer, Nasik handles UI/UX and docs, Dareen handles frontend & testing. Supervisor acts as product owner.	**Complete**
3	**Risk analysis**	Six risks rated; R1 estimation, R2 exams and R3 framework/security are key, each with a strategy, indicator and owner (see the risk register).	**Draft** (R4 to R6 not yet planned in detail)

#	Heading	One-Line Entry	Status
4	**Resource requirements**	4 members at about 6 hours a week; a Git repository and shared task board; free-tier hosting; stack confirmed by the Week 4 spike.	**Draft** (hosting limits unchecked)
5	**Work breakdown**	8 tasks (T1 to T8) totalling 28 person-days; expected range 19 to 42; 40% contingency gives about 39.	**Complete**
6	**Project schedule**	Critical path T1 → T2 or T3 → T5 → T7 → T8 = 10 weeks; milestones M1 to M3; Sprint 1 is Weeks 4 to 5.	**Needs bar chart**
7	**Monitoring and reporting**	Weekly 15-minute check on actual vs planned effort; sprint review and retrospective every 2 weeks; milestone email to the supervisor after M1, M2 and M3.	**Complete**

Action Items: Who Finishes What Before the Next Session

Task	Owner	Done By
Copy the Task 3 bar chart into the schedule section	Nasik	Next session
Check hosting limits and update the resources section	Dareen	Next session
Detail the strategies for R4 and R6	Ayoub	End of Week 5
Collect all exam dates and add them to the schedule	Zhima	End of Week 5
Merge everything into v0.2	Zhima	End of Week 5